General Mathematics — Exercise Sheet 11

Taylor approximation and small oscillations

Variant: Statements

Exercise 1 (Composition and cancellation). Use Lecture 11's Taylor formulas to solve both tasks.

1. Use Taylor's formula for the exponential to find a polynomial P of degree at most 4 satisfying the global error bound

$$0 \le P(x) - e^{-x^2} \le x^6/6 \qquad (x \in \mathbb{R}),$$

   and deduce the limit of $(e^{-x^2} - 1 + x^2)/x^4$ as $x \to 0$, $x \neq 0$.
2. Evaluate $\lim_{x\to 0}(\tan x - x)/(x - \sin x)$. Justify a deleted neighbourhood on which the quotient is defined. Use sine and cosine expansions with sufficient remainder orders.

Exercise 2 (A given relativistic energy model). Let $m, c > 0$ be fixed constants and let $v \in \mathbb{R}$ satisfy $|v| < c$. The Lorentz factor and kinetic energy in the given model are

$$\gamma(v) = \left(1 - \frac{v^2}{c^2}\right)^{-1/2}, \qquad K(v) = mc^2(\gamma(v) - 1).$$

Derive the expansions through order v^4 at zero, with error orders. Deduce the leading kinetic-energy approximation $K(v) \sim mv^2/2$ as $v \to 0$. Also prove, for $|v| \le c/2$,

$$0 \le \gamma(v) - 1 - \frac{v^2}{2c^2} \le \frac{v^4}{c^4}.$$

The physical formulas are supplied assumptions. Use the positive real root.

Exercise 3 (Irrationality from a finite remainder). Prove $e \notin \mathbb{Q}$ using the finite Taylor formula for e^x at zero and the bound $2 < e < 3$ from Lecture 4.

Exercise 4 (A strict minimum with no quadratic restoring term). In the given Newton model, let $m, \lambda > 0$ and let $V(q) = \lambda q^4/4$ on $\mathbb{R}$. Show that zero is a strict global minimum but is degenerate in Lecture 11's sense. Find the exact force and the linearised equation at zero.

For any twice differentiable motion $q : \mathbb{R} \to \mathbb{R}$ satisfying $mq'' = -\lambda q^3$ and any constant $A > 0$, define $q_A(t) = Aq(At)$. Prove that q_A satisfies the same equation. If q has least positive period P, prove that q_A has least positive period P/A. A period $P > 0$ means $q(t + P) = q(t)$ for every $t \in \mathbb{R}$.

Exercise 5 (Peano with pointwise derivatives (⋆)). This is an optional strengthening of the lecture's C^n result. Let $n \geq 1$ be an integer and let I be an open interval containing a. Suppose $f : I \to \mathbb{R}$ has derivatives through order $n-1$ on I, and $f^{(n-1)}$ is differentiable at a. The last condition supplies $f^{(n)}(a)$. Prove the Peano formula

$$f(a+h) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!} h^k + o(h^n)$$

without assuming continuity of $f^{(n)}$ near a. Use induction and the proved zero-over-zero L'Hôpital rule from Lecture 10.